

Making the PINNeAPLe Terramechanics Surrogate Robust: Auditing Physics Constraints, Hardening the Bekker–Wong Solver, and Constraint-Preserving Surrogates

Rigid-wheel / lunar-regolith interaction for rover simulation

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Abstract

PINNeAPLe ships a physics-informed surrogate of the Bekker–Wong rigid-wheel model for real-time rover simulation. We audited every physics constraint it imposes against its own reference solver and found that one of them is false: the surrogate was trained to enforce zero drawbar pull at zero slip, whereas the Bekker–Wong model gives $F_x(0, z)$ between -4.2 and 5.0 N on the preset domain. A second constraint (Mohr–Coulomb traction limit) used half the wheel circumference as contact area and was never active. We (i) corrected the preset, the compiled residual and the example, keeping only constraints verified numerically on the stated domain and adding one ($\partial F_z / \partial z \geq 0$); (ii) hardened the solver (input validation, break points at the stress kinks, sign-aware shear for braking, Brent's method for sinkage, no fabricated values on failure, and a vectorised, differentiable Gauss–Legendre quadrature that matches adaptive quadrature to 10^{-13}); and (iii) built a constraint-preserving surrogate in which the Mohr–Coulomb and torque constraints hold by construction. On an independent test set the original surrogate has relative L2 errors of 7.2% (F_x) and 22.1% (M_y); the robust surrogate 3.8% and 0.8% with zero violations of the hard constraints. At the rover's operating point the sinkage error drops from 0.72 to 0.043 mm. We also report where physics did not help (stress and extrapolation tests) and that, for this two-input problem, the batched quadrature itself (35 μ s per evaluation) is fast enough to make a neural surrogate optional.

Keywords: terramechanics; Bekker–Wong; rover; physics-informed neural networks; hard constraints; verification; PINNeAPLe; VeriPhysics

Main findings

- A physics constraint in the original PINN was wrong; enforcing it biased the model against its own data (F_x at $s = 0$ predicted ≈ 0.01 N vs mean 12.521 N from the solver).
- Constraints should be verified against the reference solver on the exact domain where they are imposed: $\partial F_x / \partial s \geq 0$ for $s \leq 0.4$ holds for $z \leq 0.058$ m (0 violations) but not up to 0.1 m (156 violations).
- Hard (architectural) constraints are cheaper and more reliable than penalties; soft penalties still leave violations.
- A data-only MLP of the same architecture is competitive on accuracy; the value of physics here is guaranteed consistency, not lower error.

1 Background

For a rigid wheel of radius R and width b at sinkage z and slip s , the Bekker–Wong model integrates normal and shear stress over the contact arc $\theta_r \leq \theta \leq \theta_f$, $\theta_f = \arccos(1 - z/R)$:

$$\sigma(\theta) = (k_c/b + k_\phi)[R(\cos \theta^* - \cos \theta_f)]^n, \tau(\theta) = (c + \sigma \tan \phi)(1 - e^{-j/K}) \quad (1)$$

$$F_x = Rb (\tau \cos \theta - \sigma \sin \theta) d\theta, F_z = Rb (\sigma \cos \theta + \tau \sin \theta) d\theta, M_y = R^2 b \tau d\theta \quad (2)$$

with shear displacement $j(\theta) = R[(\theta_f - \theta) - (1 - s)(\sin \theta_f - \sin \theta)]$ and the rear-region mapping θ^* of Wong & Reece. Soil: GRC-1 lunar simulant ($c = 1.4$ kPa, $\phi = 30^\circ$, $K = 18$ mm, $k_c = 1370$, $k_\phi = 8.14 \times 10^5$, $n = 1$); wheel $R = 0.125$ m, $b = 0.06$ m; rover 40 kg on six wheels under lunar gravity ($W = 10.8$ N per wheel).

2 Audit of the original physics constraints

Table 1. Constraint audit on a dense 4,332-point grid evaluated with the Bekker–Wong solver.

Constraint (original)	Status on preset domain ($s \leq 0.75, z \leq 0.058$ m)	Action
R1: $F_x(s=0, z) = 0$	FALSE: solver gives $-4.20 \dots 4.99$ N	removed
R2: $F_x \leq cA + F_z \tan \varphi, A = \pi bR$	holds, but never active (min margin 32.4 N)	$A = bR\theta_1$ (contact arc): min margin 1.33 N, 0 violations
R3: $\partial F_x / \partial s \geq 0, s \leq 0.4$	holds (0 violations); fails for $z \rightarrow 0.1$ m (156)	kept; domain stated
R4: $M_y \geq R F_x$	holds (min margin 0.011 N·m)	kept, made hard
R5 (new): $\partial F_z / \partial z \geq 0$	holds (0 violations)	added

The physical reason R1 fails: at $s = 0$ the shear displacement $j(\theta)$ is not zero ($\theta - \sin \theta$ is increasing), so shear stress and a net drawbar force exist; the sign depends on the balance between compaction resistance and the rear-region shear. Note also that the rover never visits most of the preset domain: for $W = 10.8$ N the sinkage stays in 7.7–8.3 mm.

3 Hardening the solver and the preset (library changes)

- **Validation:** $0 < z < R, -1 \leq s < 1$ and positive soil/wheel parameters are checked; errors raise instead of returning garbage.
- **Accurate adaptive quadrature:** the stress kinks $(\theta_m, 0)$ are passed as break points; the error estimate is exposed.
- **Batched, differentiable quadrature** (*bekker_wong_forces_torch*): piecewise Gauss–Legendre on the smooth sub-intervals, per-sample soil parameters, float64, autograd-compatible; agrees with adaptive quadrature to 10^{-13} for $n = 1$ (Figure 1).
- **Sinkage from load:** bracketed Brent root-finding on $F_z(z) = W$ (monotone, verified) replaces an unguarded secant iteration.
- **No fabricated data:** dataset generation used to write zeros for failed evaluations; it now raises or drops-and-counts.
- **Braking:** shear is sign-aware, removing an exponential blow-up for $s < 0$.
- **Preset/compiler:** false R1 condition removed; R5 added to the compiled residual; tests updated (the old test asserted the false condition) and seven new robustness tests added.

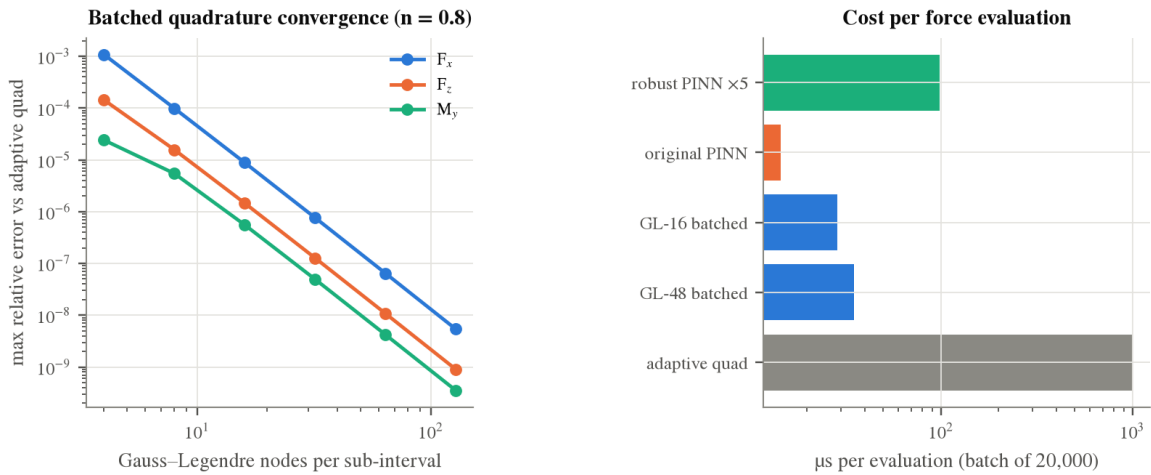


Figure 1. Left: convergence of the batched Gauss–Legendre quadrature for a non-integer sinkage exponent ($n = 0.8$), where the integrand is not smooth at θ_1 . Right: cost per evaluation.

4 Surrogates

- **Original:** the frozen example (git HEAD) run unchanged — Fourier-feature ResNet, 4,000 epochs, R1–R4 penalties, its own grid+random data (2,531 points, random 85/15 split, no test set).
- **Data-only MLP:** PINNeAPPLE ModifiedMLP, same inputs, no physics; 5-member deep ensemble.

- **Robust PINN** (this work): same backbone; outputs parameterised so that $F_z > 0$, $F_x \leq c_b R \theta_1 + F_z \tan \phi$ and $M_y \geq R F_x$ hold by construction; soft penalties for the monotonicity constraints R3, R5 on unlabelled collocation points; load-relative loss; 5-member deep ensemble with variance recalibrated on validation.
- Data: 2,000 training and 300 validation Latin-hypercube points; an independent 5,000-point Latin-hypercube test set (seed not used anywhere else).

5 Results

Table 2. Independent test set (5,000 points). Violation rates on 20,000 fresh points (finite differences for R3, R5).

Model	rel. L2 F_x	rel. L2 F_z	rel. L2 M_y	R2 viol.	R3 viol.	R4 viol.	R5 viol.
Original PINN	7.21%	1.27%	22.08%	0.00%	2.54%	0.00%	0.07%
Data-only MLP	2.20%	1.51%	1.81%	0.00%	0.62%	0.07%	2.76%
Robust PINN (this work)	3.76%	0.70%	0.79%	0.00%	0.00%	0.00%	0.73%

Table 3. Rover operating point: sinkage from $F_z(s, z) = 10.8$ N solved with each surrogate, and the drawbar pull there.

Model	Sinkage RMSE (mm)	Drawbar-pull RMSE (N)
Original PINN	0.716	1.602
Data-only MLP (Modified MLP)	0.015	0.018
Robust PINN (hard R2,R4 + soft R3,R5)	0.043	0.035

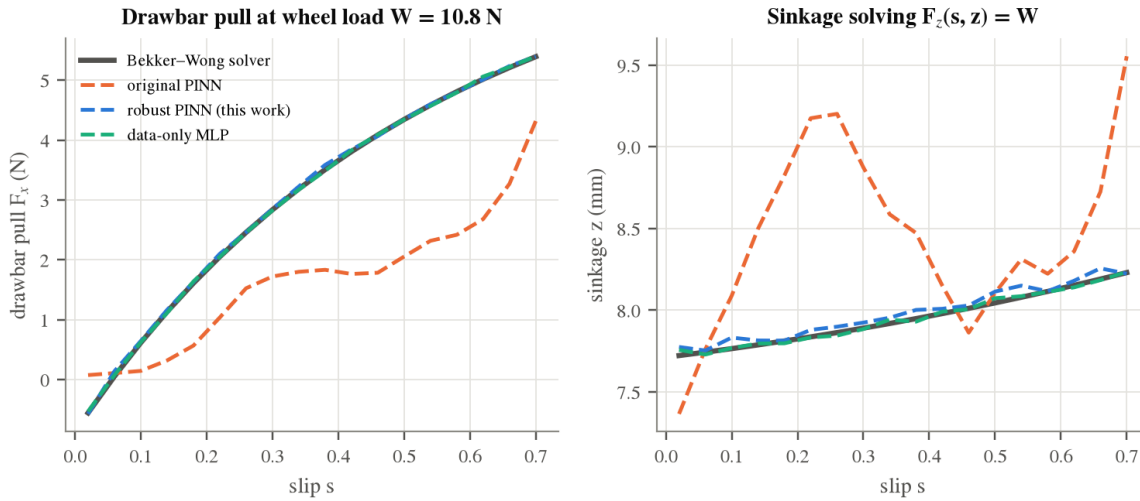


Figure 2. Operating-point behaviour of the surrogates against the Bekker–Wong solver.

Uncertainty: the robust ensemble (variance scaled on validation) reaches 90%-interval coverage of 0.901 (F_x), 0.904 (F_z), 0.931 (M_y) on test, with ECE 0.050, 0.044, 0.051.

5.1 Stress tests (equal gradient-step budget)

Table 4. Label noise (3%), small data (150 points) and extrapolation in sinkage (train $z \leq 0.04$ m, test $z > 0.045$ m). 3-member ensembles, $\geq 2,400$ gradient steps each.

Case	Model	rel. L2 F_x	rel. L2 F_z	rel. L2 M_y	R3 viol.	R5 viol.
label noise 3pct	Data-only MLP	2.7%	2.1%	2.3%	1.54%	3.00%
label noise 3pct	Robust PINN	4.9%	1.6%	1.4%	0.00%	0.82%
small data 150	Data-only MLP	54.3%	54.7%	40.5%	38.42%	24.30%
small data 150	Robust PINN	51.0%	34.9%	29.6%	2.38%	10.92%
extrapolate sinkage train $z \leq 0.04$ test $z > 0.045$	Data-only MLP	80.5%	85.5%	88.2%	13.23%	30.44%
extrapolate sinkage train $z \leq 0.04$ test $z > 0.045$	Robust PINN	75.8%	79.9%	75.7%	11.31%	28.92%

Physics constraints bound the answer (they cannot be violated) but do not supply the magnitudes that are missing from the data: under extrapolation both models degrade by an order of magnitude. This is the expected limit of inequality constraints.

5.2 Parametric surrogate across soils

Table 5. 8-input surrogate (slip, sinkage, c , ϕ , K , k_c , k_ϕ , n): 30,000 training points generated in 4 s with the batched quadrature (max relative deviation from adaptive quadrature $3.2e-08$). OOD: every soil parameter beyond the training box. AUROC: detection of OOD inputs from the ensemble spread.

Model	in-dist. F_x	in-dist. F_z	in-dist. M_y	OOD F_x	OOD F_z	OOD M_y	OOD AUROC
Data-only MLP (linear outputs)	4.6%	1.6%	2.4%	82.7%	75.8%	79.4%	1.000
Data-only MLP (physically scaled outputs)	0.9%	0.4%	0.6%	67.2%	73.0%	77.2%	1.000
Robust PINN	1.8%	0.7%	0.8%	37.3%	67.7%	43.1%	1.000

5.3 VeriPhysics validation

Table 6. PINNeAPLe PhysicsGuardrail + compute_physics_confidence. numerical_convergence is the GCI of the reference quadrature (observed order 3.49, GCI $7.3e-07$).

Model	Trust score	Coverage	Guardrail checks (pass/fail)	Components
Robust PINN (hard R2,R4 + soft R3,R5)	0.902	0.8	parameter_sanity=P, pde_residual=F, reference_data_match=P	physics_guardrail 0.67, numerical_convergence 1.00, uq_calibration 0.95, benchmark_agreement 0.99
Data-only MLP (ModifiedMLP)	0.809	0.8	parameter_sanity=P, pde_residual=F, reference_data_match=F	physics_guardrail 0.33, numerical_convergence 1.00, uq_calibration 0.92, benchmark_agreement 0.98
Original PINN	0.771	0.6	parameter_sanity=P, pde_residual=F, reference_data_match=F	physics_guardrail 0.33, numerical_convergence 1.00, benchmark_agreement 0.98

All three models fail the guardrail's pde_residual check. This is informative about the check, not only the models: for this problem kind the compiled residual is dimensional (N^2 for the inequality violations, N^2/m^2 for $\partial F_z / \partial z$), so its default threshold of 10^{-2} has no physical meaning. The dimensionless violation rates of Table 2 are the meaningful physics metric; normalising the compiled terramechanics residual is recorded as follow-up work for the library.

6 Conclusions

- Audit physics constraints against the reference model before imposing them; one of four was false and one was vacuous.
- Prefer hard, architectural constraints where the physics gives inequalities with closed forms; they cost nothing at inference.
- Report data-only baselines: here they match physics-informed accuracy, and the benefit of physics is consistency.
- For low-dimensional models, a batched, verified numerical method can be the right 'surrogate'.

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